CHAPTER 10

Trigonometry: circle, radians and angles Associates

Chapter overview

Objectives. Switch degrees to radians; place dots; calculate sine and cosine remarkable; exploit the symmetries of the circle.
Prerequisites. Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.
Annual benchmark. 2·11 March: 8 a.m.

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Perimeter of a circle of radius 1 ?

2. How many degrees is a quarter of a turn worth?

3. Calculate $\sqrt{\frac{1}{2}}$.

4. Abscess and ordinate of the symmetrical of $(a; b)$ in relation to the axis of the apses?

5. Simplify $\frac{5\pi}{3} - 2\pi$.

6. In a right triangle, opposite side 3 and hypotenuse 5: sine?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Present an associated equality of angles by naming the symmetry that justifies it; explain what a radian measures.

Search before formalizing

Build, without rapporteur, a corner point $\frac{\pi}{3}$ then a corner point $\frac{\pi}{4}$. Justify the exact values their coordinates from the figures.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

10.1. Measure an angle in radian

Definition and results

The trigonometric circle is the center circle O and radius 1, oriented in the inverse direction
A radian corresponds to an arc of length 1 on this circle.
value 2π radians, half a turn π. On a radius circle r, a length of arc is $r\theta$ for one positive angle θ expressed in radians.

Status and justification. Definitions and proportionality between centre angle and arc length.
Conversion is $180^\circ = \pi$ radians; multiplies the degrees by $\frac{\pi}{180}$.

Example solved. $60^\circ = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270^\circ$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of $2\ \pi$ Inch.

Point of vigilance

The length of an arc is not that of the rope. In trigonometric calculation, check the mode degree or radian of tool.

Immediate application. Convert 45°, 120° ; give the radius arc 2 m and angle $\frac{3\pi}{4}$.

10.2 Winding and image of a real

Definition and results

To each real x corresponds an M point of the circle obtained by a length oriented course x since I(1;0). Two reals have the same image if their difference is an integer multiple of 2 π.
A negative number indicates a route in the clockwise direction.

Status and justification. Adding or removing an integer number of turns keeps the point of arrival. Conversely, returning to the same point requires an integer number of turns.

Example solved. $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: point $(0; -1)$.

Point of vigilance

Add π gives the opposite point, not the same point.

Immediate application. Place images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ Bringing them back to a turn.

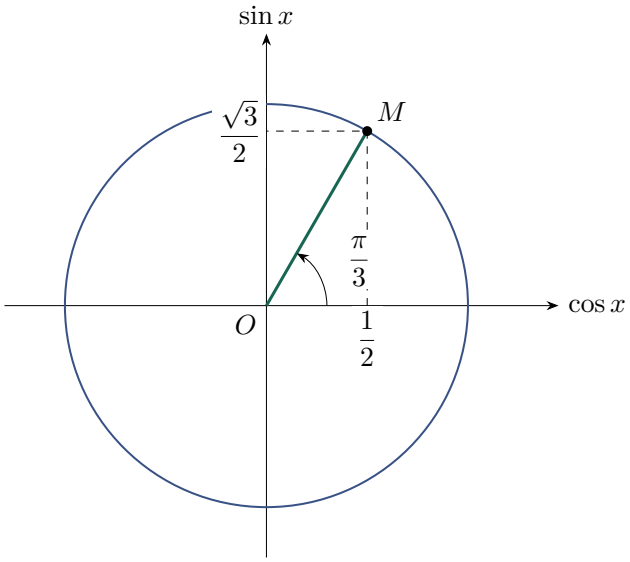
10.3 Sine, Cosine and Remarkable Values

Definition and results

If M is the image of x, $M = (\cos x; \sin x)$ Therefore $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$.
For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosine are 1 $, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}$, 0 and sine 0 $, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Status and justification. À $\frac{\pi}{4}$, the two coordinates are equal and positive ; Pythagoras gives $2a^2 = 1$, therefore $a = \frac{\sqrt{2}}{2}$. An equilateral triangle on side 1, cut by its height, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. Additional angle gives $\frac{\pi}{6}$.

Example solved. The angle point $\frac{\pi}{3}$ is $(\frac{1}{2}; \frac{\sqrt{3}}{2})$ The scalar product of two unit vectors making this angle is worth its cosine, so $\frac{1}{2}$. The geometry chapters thus connect to the circle.



For angle $\frac{\pi}{3}$, the projections of M give directly its cosine and sine.
placed outside the lines of the drawing.

Point of vigilance

The cosine is the abscisse, the sine is the ordinate. The identity gives an absolute value: the quadrant determines the sign.

Immediate application. If $\cos x = \frac{3}{5}$ and image of x is in the fourth quadrant, giving $\sin x$.

10.4 Associated angles and symmetries

Definition and results

The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Add 2 $k\pi$, k integer, does not change no sine or cosine.

Status and justification. Each equality is read on the coordinates of the symmetrical points: axis horizontal for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a catalogue Without justification.

Example solved. $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry of $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Point of vigilance

Two angles of the same cosine do not necessarily have the same sine.
read the signs.

Immediate application. Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

10.5 Approximate pi and reinvest

Definition and results

A regular polygon inscribed n sideways in the half-perimeter unit circle $n\sin(\frac{\pi}{n})$; a polygone circumscribed to half-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ Here. Archimedes method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi In advance.

Status and justification. The marked perimeter is lower than that of the circle and circumscribes it top: geometric frame allowed. If it is the rope on one side, place $h = \sqrt{1 - \frac{c^2}{4}}$; the rope of half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids subtraction of real close.

Example solved. For the listed hexagon, c=1 and half-perimeter 3. $\sqrt{3} \approx 3{,}4641$. By doubling the sides, the bounds is tightened. The programme provided starts from n=6 and c=1.

Point of vigilance

Directly using a sine function with already known pi would check a calculation, but does
Would not build an independent pi approximation.

Immediate application. For n=6 and c=1, calculate the new string after doubling.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 10.1 · Convert and Measure

The trigonometric circle is used to measure radian angles.

1. Convert 150 $^\circ$ and -90° in radians, showing the conversion factor.
2. A circular sector has a radius of 4 cm and an angle of $\frac{\pi}{3}$ radians. Calculate the exact length of his bow.
3. Explain why a angle measurement can be negative while a geometric arc length is positive.

Exercise 10.2 · Reduce angles

We consider the real $\frac{17\pi}{6}$, $-\frac{7\pi}{4}$ and 5π.

1. For each, determine a representative in [0; 2 π[having the same image on the circle Trigonometric. Show the number of turns added or removed.
2. Place the three images on a circle and identify them by the actual starting.

Exercise 10.3 · Accurate values

The exact values of the cosine and sine of $\frac{2\pi}{3}$, $-\frac{\pi}{6}$ and $\frac{5\pi}{4}$.

1. Place the three images on the trigonometric circle. For each, specify the expected sign of its two coordinates.
2. Calculate the exact values using a remarkable angle and symmetry. answer in a table: angle, cosine, sine.

Exercise 10.4 · Find the sign

A real one x checks $\sin x = \frac{5}{13}$. Its image on the trigonometric circle is located in the second quadrant, therefore to the left of the axis of the y-axis and above the axis of the abscisses.

1. Determine the exact value of cos x. Justify both its absolute value and its sign.

2. Explain why the only sin data x would not have been enough to choose the sign of cos x.

Exercise 10.5 · One wheel

One wheel of radius 0,30 m rolls on a horizontal ground without slipping. It rotates angle $\frac{5\pi}{3}$ The distance travelled by its axis in the direction of displacement is sought.

1. Calculate this distance in exact form by explaining the link with the length of the arc.

2. Explain why one should not use the length of a rope in the circle.
fixed point on the edge of the wheel, seen from the ground, is it that of the axis?

Exercise 10.6 · String and scalar product

On the centre trigonometric circle O, point A is the image of 0 and the point B that of $\frac{2\pi}{3}$.

1. Determine the exact coordinates of A and B, then calculate the length AB.

2. Retrieve this length by applying the relationship of Al-Kashi to the triangle OAB. Specify the lengths and angle used.

Exercise 10.7 · Read an equation on the circle

We're looking for real things. x of the interval [0; 2 π[whose image on the trigonometric circle has for abscisse $\frac{1}{2}$.

1. Translate the condition on the abscisse by an equation involving cos x.

2. Trace the corresponding vertical line on a circle, identify its intersections and give the
Two solutions. Check that they belong to the required interval.

Fiscal year 10.8 · Deferred review

The measuring angle shall be considered $\frac{25\pi}{4}$ The answers must be accurate, without approximation. decimal places.

1. Reduce this angle by showing an integer number of complete turns.

2. Deduce its cosine and sine. Justify the values using the trigonometric circle.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 32 p. 220, questions a and b. Use the exact value provided and angles partners, without calculator.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. Addressing the four issues of calculation and reading the circle; justify the answers on the copy.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $180°$ value
   - A. 2π
   - B. π
   - C. $\frac{\pi}{2}$
   - D. 1
2. $\cos(\frac{\pi}{3}) =$
   - A. $\frac{1}{2}$
   - B. $\frac{\sqrt{3}}{2}$
   - C. 0
   - D. 1
3. $\sin(-\frac{\pi}{2}) =$
   - A. 1
   - B. 0
   - C. $\frac{1}{2}$
   - D. -1
4. $\cos^2 x + \sin^2 x =$
   - A. 0
   - B. 2
   - C. 1
   - D. x
5. The same point corresponds to angles different from
   - A. π
   - B. $2k\pi$, k integer
   - C. $\frac{\pi}{2}$
   - D. 1

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $\cos(\pi - x) =$

   A. $\cos x$

   B. $\sin x$

   C. $-\cos x$

   D. $-\sin x$

2. $\sin(\frac{\pi}{4}) =$

   A. $\frac{\sqrt{2}}{2}$

   B. $\frac{1}{2}$

   C. $\frac{\sqrt{3}}{2}$

   D. $\sqrt{2}$

3. Image of $\frac{3\pi}{2}$:

   A. $(1;0)$

   B. $(-1;0)$

   C. $(0;1)$

   D. $(0;-1)$

4. Radius arc 3 and angle 2 radians: length

   A. $\frac{2}{3}$

   B. 6

   C. $\frac{3}{2}$

   D. 6π

5. Interior of the second quadrant: sign of the cosine

   A. Positive

   B. None anywhere

   C. Negative

   D. Indeterminable

True or false?

1. $\cos(-x) = -\cos x$ for all x.

2. $\sin(x + 2\pi) = \sin x$.

3. The data of sin x determines always cos x.

Personal review

I reduce the angles by complete turns; I read cosine and sine in the right order; I justify the signs;
I keep the exact values.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and
a mistake to avoid.

Corrected and conduct of chapter 10

Use as a teacher. The following answers relate to the outstanding tasks in the course. first clarify the approach; provide graduated help before the solution. estimates to be calibrated in class.

Diagnosis and Remediation

1. 2π. *Action:* Connect radius, diameter and perimeter.
2. 90°. *Action:* Share a circle in four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Streamline and check square.
4. $(a;-b)$. *Action:* Draw both points.
5. $-\dfrac{\pi}{3}$. *Action:* Put at the same denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

Research activity

Equilateral triangle then isosceles right triangle. Pythagoras gives respectively height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

Immediate applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length 3 $\dfrac{\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ maps to the value $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $(-\dfrac{1}{2};-\dfrac{\sqrt{3}}{2})$ and $(\dfrac{\sqrt{2}}{2};-\dfrac{\sqrt{2}}{2})$.
5. $h = \dfrac{\sqrt{3}}{2}$; new rope $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Exercises: approach and results

Exercise 10.1 · Convert and measure. $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates a orientation; geometric length is positive.

Exercise 10.2 · Reduce angles. $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, π. Remove or add multiples of 2 π.

Exercise 10.3 · Accurate values. $(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2};-\frac{1}{2})$; $(-\frac{\sqrt{2}}{2};-\frac{\sqrt{2}}{2})$ in form (cosine ; sine).

Exercise 10.4 · Find the sign. $\cos^2 x = 1-\frac{25}{169}=\frac{144}{169}$; negative cosine, therefore $-\frac{12}{13}$.

Exercise 10.5 · One wheel. Progress $r\theta = 0{,}3(\frac{5\pi}{3})=\frac{\pi}{2}$ m. This corresponds to the course of the arc; the rope connects two positions on the circle and does not have that length. The material point on the wheel describes a different trajectory from the ground.

Exercise 10.6 · String and scalar product. $A(1;0)$, $B(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $AB^2=\frac{9}{4}+\frac{3}{4}$ = 3. Al-Kashi : $1+1-2\cos(\frac{2\pi}{3})$ = 3. $AB=\sqrt{3}$.

Exercise 10.7 · Read an equation on the circle. The vertical abscisse $\frac{1}{2}$ cuts the circle to the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. Both points have opposite sinuses.

Exercise 10.8 · Deferred review. $\frac{25\pi}{4}=6\pi+\frac{\pi}{4}$. Three full turns do not change the image, So the two values are worth $\frac{\sqrt{2}}{2}$.

CQFD, exercise 32 p. 220, questions a and b. a : $\cos(\frac{6\pi}{5})=\cos(\pi+\frac{\pi}{5})=-\frac{1+\sqrt{5}}{4}$. b : $\cos(-\frac{\pi}{5})=\cos(\frac{\pi}{5})=\frac{1+\sqrt{5}}{4}$, by cosine parity.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. 1 : $\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$. 2 : yes, $\cos(\frac{\pi}{3})=\frac{1}{2}=0{,}5$ is decimal. 3: the main measure in] $-\pi;\pi$] is $-\frac{5\pi}{6}$. 5 : $\sin(\frac{4\pi}{3})=-\frac{\sqrt{3}}{2}$.

MCQ: answers and reasons

MCQ 1.

1. B. Half-turn.

2. A. Equilateral triangle.

3. D. Lower point of the circle.

4. C. Equation of the unity circle.

5. B. Complete towers.

MCQ 2.

1. C. Vertical Symmetry.

2. A. Isosceles right triangle.

3. D. Three-quarters of a turn.

4. B. $r\theta = 6$.

5. C. Negative abscess inside the quadrant.

True or false

1. False: in 0, the two members would be 1 and -1. Cosine is even.

2. True: a complete tower preserves the image.

3. False: $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ have even sine but cosine opposites.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.